

COMPUTER NETWORKS — COMPLETE NOTES

Covers: College Exams | Placement Preparation | Technical Interviews | Quick Revision

SECTION 1 — NETWORKING FUNDAMENTALS

1. Computer Network

Definition

A **computer network** is a collection of **two or more devices (computers, servers, phones)** connected together to **share data and resources**.

Need / Purpose

- **Resource Sharing** → Share printers, files, internet connections.
- **Communication** → Email, messaging, video calls.
- **Data Storage** → Centralized servers; remote storage.
- **Cost Reduction** → Share expensive hardware across users.

Types of Networks

Type	Full Form	Coverage	Example
PAN	Personal Area Network	~10 meters	Bluetooth, USB
LAN	Local Area Network	Building / Campus	Office Wi-Fi, Ethernet
MAN	Metropolitan Area Network	City	Cable TV network

WAN	Wide Area Network	Country / World	Internet, leased lines
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In One Line

A computer network connects devices to share data, resources, and services.

2. Internet

Definition

The **Internet** is a **global network of interconnected networks** that uses the **TCP/IP protocol suite** to communicate.

Key Points

- Made up of millions of private, public, academic, and government networks.
- Data travels as **packets** across routers.
- **WWW (World Wide Web)** is a service on the Internet — not the Internet itself.
- Governed by standards from **IETF, ICANN, IEEE**.

In One Line

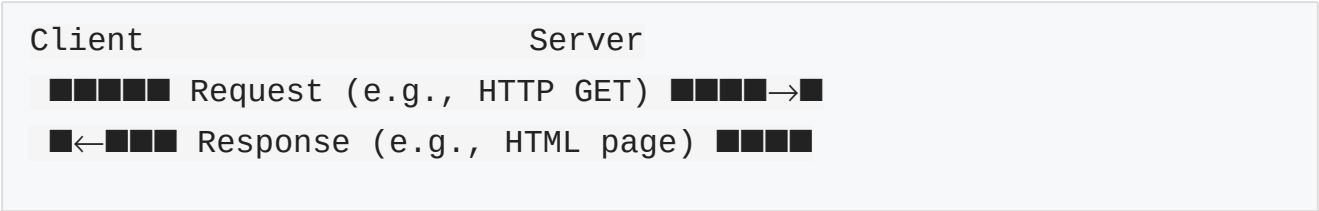
The Internet is the **global network of networks communicating via TCP/IP**.

3. Client and Server

Definition

- **Client** → A device or program that **requests** a service or resource.
- **Server** → A device or program that **provides** a service or resource.

Client-Server Model



Key Points

- **Client** initiates communication; **Server** listens and responds.
- A single machine can act as both client and server.
- **P2P (Peer-to-Peer)** → Every node acts as both client and server. Example: BitTorrent.

In One Line

Client requests, server responds — the fundamental model of networked communication.

4. Network Protocol

Definition

A **network protocol** is a **set of rules and conventions** that define how data is **formatted, transmitted, received, and acknowledged** between devices.

Why Needed

- Without protocols, devices from different manufacturers cannot communicate.
- Protocols define **format, timing, sequencing, and error handling**.

Examples

Protocol	Purpose
HTTP/HTTPS	Web browsing
TCP	Reliable data transfer

- Smaller packets → Better error detection; less retransmission overhead.

In One Line

A packet is a small chunk of data with a header (addressing info) and payload (actual data) sent across a network.

6. Port

Definition

A **port** is a **logical endpoint** for communication, used to identify a specific **process or service** on a host. Identified by a 16-bit number (0–65535).

Port Ranges

Range	Type	Description
0–1023	Well-known ports	Reserved for standard services
1024–49151	Registered ports	Assigned by IANA to applications
49152–65535	Dynamic/Ephemeral ports	Used by clients temporarily

Common Port Numbers

Port	Protocol
20, 21	FTP
22	SSH

23	Telnet
25	SMTP
53	DNS
80	HTTP
443	HTTPS
3306	MySQL
3389	RDP

In One Line

A port is a logical endpoint that identifies a specific service on a host — IP identifies the host, port identifies the service.

7. Socket

Definition

A **socket** is an **endpoint for communication** between two programs over a network, identified by the combination of **IP address + Port number + Protocol**.

Socket = IP Address + Port + Protocol

Socket: 192.168.1.10 : 443 : TCP

■■■■■■■■■■■■■■■■■■■■ ■■■ ■■■■

IP Address Port Protocol

Key Points

- A **connection** is uniquely identified by a pair of sockets (client socket + server socket).

- **Stream socket (TCP)** → Reliable, connection-oriented.
- **Datagram socket (UDP)** → Unreliable, connectionless.
- In programming: `socket()`, `bind()`, `listen()`, `accept()`, `connect()`, `send()`, `recv()`.

In One Line

A socket = IP + Port + Protocol — the unique endpoint through which two processes communicate.

8. IP Address

Definition

An **IP (Internet Protocol) address** is a **unique numerical label** assigned to every device on a network, used for **identification and routing**.

IPv4 vs IPv6

Feature	IPv4	IPv6
Format	32-bit, dotted decimal (e.g., 192.168.1.1)	128-bit, hexadecimal (e.g., 2001:db8::1)
Total addresses	~4.3 billion	$\sim 3.4 \times 10^3$ ■
Notation	4 octets (0–255)	8 groups of 4 hex digits
Header size	20–60 bytes	Fixed 40 bytes
NAT needed	Yes (address exhaustion)	No

In One Line

An IP address uniquely identifies a device on a network; IPv4 is 32-bit, IPv6 is 128-bit.

9. MAC Address

Definition

A **MAC (Media Access Control) address** is a **hardware address** permanently assigned to a network interface card (NIC) by the manufacturer. Used for communication within the same network (Layer 2).

Format

AA:BB:CC:DD:EE:FF (6 bytes = 48 bits, hexadecimal)

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OUI (manufacturer) Device-specific

MAC vs IP

Feature	MAC Address	IP Address
Layer	Data Link (Layer 2)	Network (Layer 3)
Scope	Local network (LAN)	Global (Internet)
Assigned by	Manufacturer (NIC)	Network admin / DHCP
Changes	No (burned-in)	Yes (dynamic)
Format	48-bit hex	32-bit (IPv4) or 128-bit (IPv6)

In One Line

MAC address is a permanent hardware address used for local network communication; IP is used for routing across networks.

10. Bandwidth

Definition

Bandwidth is the **maximum amount of data** that can be transmitted over a network connection in a given time period. Measured in **bps (bits per second)**.

Key Terms

- **Bandwidth** → Maximum theoretical capacity (e.g., 100 Mbps Ethernet).
- **Throughput** → Actual data rate achieved in practice (always $\leq$ bandwidth).
- **Goodput** → Useful application-level throughput (excluding protocol overhead).

Units

Unit	Value
1 Kbps	1,000 bits/sec
1 Mbps	1,000,000 bits/sec
1 Gbps	1,000,000,000 bits/sec

In One Line

Bandwidth is the maximum data capacity of a link; throughput is what you actually get.

11. Latency

Definition

Latency is the **time delay** between sending data and receiving it at the destination. Also called **delay** or **ping**.

Components of Latency

- **Propagation delay** → Time for signal to travel the physical medium (speed of light).
- **Transmission delay** → Time to push all packet bits onto the wire = Packet size / Bandwidth.
- **Processing delay** → Time routers take to process packet headers.
- **Queuing delay** → Time packet waits in router queue.

Total Latency = Propagation + Transmission + Processing + Queuing

In One Line

Latency is the total delay for data to travel from source to destination — lower is better.

SECTION 2 — NETWORK MODELS

12. OSI Model

Definition

The **OSI (Open Systems Interconnection) Model** is a **conceptual 7-layer framework** that standardizes how different network systems communicate.

The 7 Layers

Layer 7 – Application	(HTTP, FTP, SMTP, DNS)
Layer 6 – Presentation	(Encryption, Compression, Encoding)
Layer 5 – Session	(Session management, Authentication)
Layer 4 – Transport	(TCP, UDP – end-to-end communication)
Layer 3 – Network	(IP – routing, logical addressing)
Layer 2 – Data Link	(MAC – frames, local delivery, error detection)
Layer 1 – Physical	(Cables, signals, bits)

Mnemonic: "All People Seem To Need Data Processing" (top to bottom)

Layer Functions

Layer	Name	PDU	Key Protocols	Job
7	Application	Data	HTTP, FTP, DNS, SMTP	User interface; application services
6	Presentation	Data	SSL/TLS, JPEG, MPEG	Format, encrypt, compress data
5	Session	Data	NetBIOS, RPC	Establish, manage, terminate sessions
4	Transport	Segment	TCP, UDP	End-to-end delivery, reliability
3	Network	Packet	IP, ICMP, OSPF	Logical addressing, routing
2	Data Link	Frame	Ethernet, ARP, PPP	Physical addressing (MAC), framing

1	Physical	Bit	Ethernet cable, Wi-Fi	Transmit raw bits over medium
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In One Line

OSI Model is a 7-layer reference framework: **Physical** → **Data Link** → **Network** → **Transport** → **Session** → **Presentation** → **Application**.

13. Physical Layer

Definition

The **Physical Layer (Layer 1)** is responsible for **transmitting raw bits** over a physical medium (cable, fiber, wireless).

Key Points

- Deals with **electrical signals, light pulses, or radio waves**.
- Defines: connector types, pin layout, cable type, voltage levels, data rates.
- **PDU**: Bit.
- Examples: Ethernet cables (Cat5e, Cat6), fiber optic, Wi-Fi radio signals, hubs.
- Devices: Hub, Repeater, Modem.

In One Line

Physical layer transmits raw bits — it's the hardware and signal level of networking.

14. Data Link Layer

Definition

The **Data Link Layer (Layer 2)** is responsible for **node-to-node delivery** of frames within the same network, using **MAC addresses**.

Key Functions

- **Framing** → Wraps packets into frames.
- **Physical Addressing** → Uses MAC address for local delivery.
- **Error Detection** → CRC (Cyclic Redundancy Check) in frame trailer.
- **Flow Control** → Manages data rate between sender and receiver.
- **Access Control** → MAC protocols (CSMA/CD for Ethernet).

Sub-layers

- **LLC (Logical Link Control)** → Error control, flow control.
- **MAC (Media Access Control)** → Physical addressing, media access.

Devices: Switch, Bridge.

In One Line

Data Link layer delivers frames within a LAN using MAC addresses and detects errors with CRC.

15. Network Layer

Definition

The **Network Layer (Layer 3)** is responsible for **logical addressing (IP)** and **routing** packets from source to destination across multiple networks.

Key Functions

- **Logical Addressing** → Assigns and uses IP addresses.
- **Routing** → Determines best path for packets.
- **Packet Forwarding** → Moves packets hop-by-hop toward destination.
- **Fragmentation** → Splits packets if too large for the next network's MTU.

Key Protocols

- **IP (IPv4/IPv6)** → Addressing and delivery.
- **ICMP** → Error reporting (used by ping).
- **OSPF, BGP, RIP** → Routing protocols.

Devices: Router, Layer 3 Switch.

In One Line

Network layer handles IP addressing and routing — getting packets from source to destination across networks.

16. Transport Layer

Definition

The **Transport Layer (Layer 4)** provides **end-to-end communication** between applications on different hosts, handling reliability, flow control, and multiplexing.

Key Functions

- **Segmentation** → Breaks data into segments.
- **Port-based Multiplexing** → Multiple apps use network simultaneously via ports.
- **Reliability (TCP)** → Acknowledgements, retransmission, sequencing.
- **Flow Control** → Prevents sender from overwhelming receiver.
- **Congestion Control** → Prevents network overload.

Key Protocols: TCP (reliable), UDP (fast).

In One Line

Transport layer delivers data end-to-end between applications using ports, with TCP (reliable) or UDP (fast).

17. Application Layer

Definition

The **Application Layer (Layer 7)** provides **network services directly to user applications** — the layer humans interact with.

Key Protocols

Protocol	Purpose
HTTP/HTTPS	Web browsing
DNS	Domain name resolution
SMTP/IMAP/POP3	Email
FTP/SFTP	File transfer
SSH	Secure remote login
DHCP	IP address assignment
Telnet	Unsecure remote login

In One Line

Application layer is where user-facing protocols live — HTTP, DNS, FTP, SMTP, SSH.

18. TCP/IP Model

Definition

The **TCP/IP Model** (also called the **Internet Model**) is the **practical 4-layer model** used by the actual Internet.

The 4 Layers

Layer 4 – Application	(HTTP, FTP, DNS, SMTP, SSH)
Layer 3 – Transport	(TCP, UDP)
Layer 2 – Internet	(IP, ICMP, ARP)
Layer 1 – Network Access	(Ethernet, Wi-Fi, physical hardware)

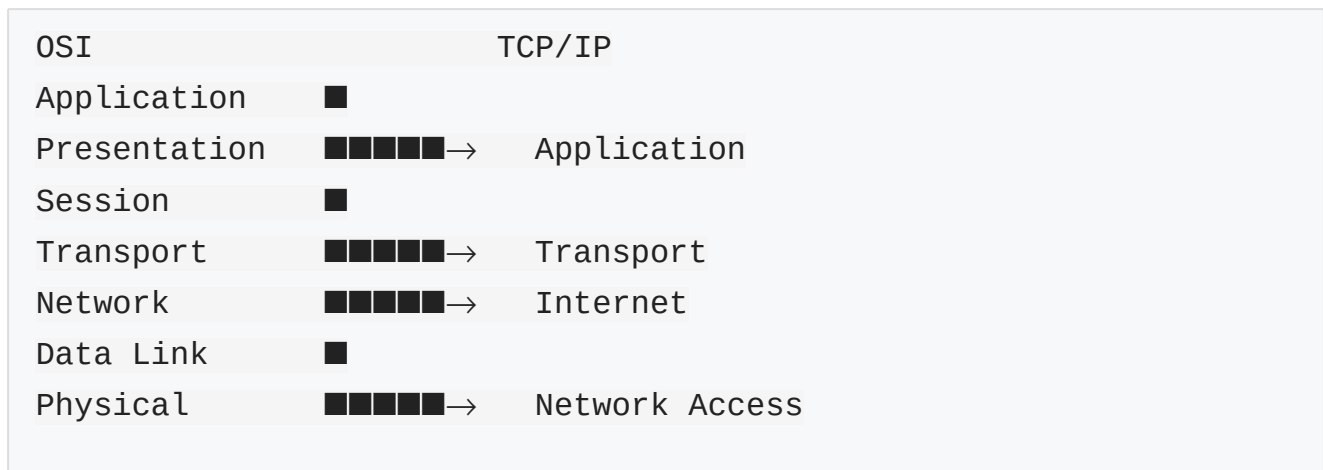
In One Line

TCP/IP is the 4-layer practical model that powers the Internet: Network Access
→ Internet → Transport → Application.

19. OSI vs TCP/IP

Feature	OSI Model	TCP/IP Model
Layers	7	4
Purpose	Conceptual reference	Practical implementation
Usage	Teaching, troubleshooting	Actual Internet
Transport protocols	Defined generically	TCP and UDP
Application layers	Session + Presentation + Application	Combined into Application
Developed by	ISO	ARPANET / DoD
Protocol independence	Protocol-independent model	Built around TCP/IP

Mapping



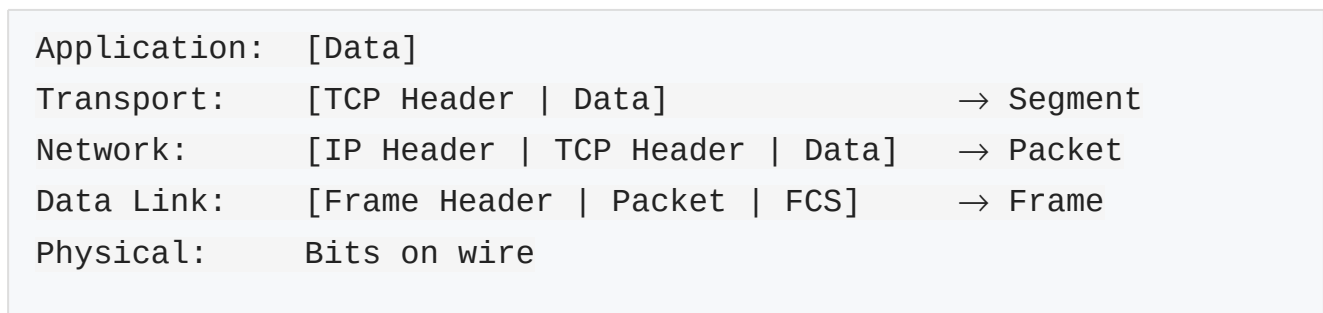
In One Line

OSI = 7-layer reference model; TCP/IP = 4-layer practical model; TCP/IP collapses OSI's top 3 and bottom 2 layers.

20. Encapsulation & Decapsulation

Encapsulation

Adding headers (and trailers) at each layer as data travels down the sender's protocol stack.



Decapsulation

Removing headers at each layer as data travels up the receiver's protocol stack — reverse of encapsulation.

In One Line

Encapsulation wraps data with headers at each layer going down; decapsulation unwraps them going up.

SECTION 3 — DATA LINK & LOCAL NETWORKING

21. MAC Address (Detail)

Definition

A **MAC (Media Access Control) address** is a **48-bit (6-byte) unique hardware address** burned into every NIC, used for local network communication.

Key Points

- First 3 bytes = **OUI** (Organizationally Unique Identifier — manufacturer ID).
- Last 3 bytes = Device-specific identifier.
- Written as: AA:BB:CC:DD:EE:FF or AA-BB-CC-DD-EE-FF.
- **ARP** maps IP addresses to MAC addresses on a LAN.

In One Line

MAC address = unique 48-bit NIC hardware address used for local delivery within a LAN.

22. Ethernet

Definition

Ethernet is the most widely used **wired LAN technology** (IEEE 802.3 standard) that defines how data is transmitted over cables using frames.

Key Points

- Uses **CSMA/CD** (Carrier Sense Multiple Access with Collision Detection) for media access.
- Data sent as **frames** with source/destination MAC addresses.
- Speeds: 10 Mbps → 100 Mbps → 1 Gbps → 10 Gbps → 100 Gbps.

- Cable types: Cat5e (100m, 1 Gbps), Cat6 (100m, 10 Gbps), Fiber optic.

In One Line

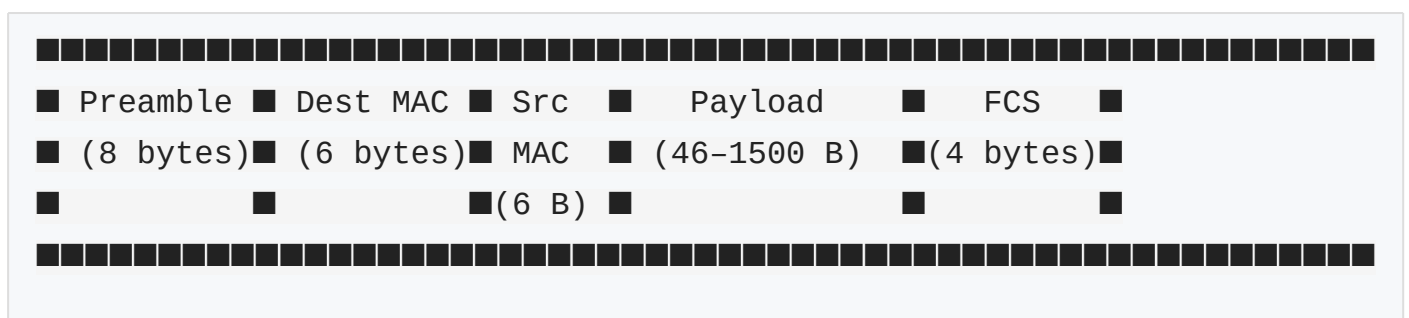
Ethernet is the standard wired LAN technology using frames, MAC addresses, and CSMA/CD.

23. Frame

Definition

A **frame** is the **Layer 2 PDU (Protocol Data Unit)** — a packet of data with a header containing MAC addresses and a trailer with error detection.

Ethernet Frame Structure



- **Preamble** → Synchronization.
- **Destination / Source MAC** → Who it's for / from.
- **Payload** → Encapsulated packet (IP packet).
- **FCS (Frame Check Sequence)** → CRC for error detection.

In One Line

A frame is the Layer 2 data unit with source/destination MAC addresses and CRC error check.

24. ARP (Address Resolution Protocol)

Definition

ARP is a protocol that **resolves an IP address to a MAC address** within a local network (LAN).

How ARP Works

1. Host A wants to send data to IP 192.168.1.5
2. A checks ARP cache – not found
3. A broadcasts ARP Request: "Who has 192.168.1.5?"
4. Host B (192.168.1.5) replies with its MAC address (unicast)
5. A updates its ARP cache and sends the frame

Key Terms

- **ARP Cache** → Local table storing IP→MAC mappings (temporary).
- **ARP Request** → Broadcast asking for MAC of a given IP.
- **ARP Reply** → Unicast response with the MAC address.
- **Gratuitous ARP** → A host announces its own IP→MAC mapping.

In One Line

ARP maps IP addresses to MAC addresses so that frames can be delivered correctly within a LAN.

25. Hub

Definition

A **hub** is a Layer 1 networking device that **broadcasts all incoming data to every connected device**, regardless of the destination.

Key Points

- **Dumb device** — no intelligence; no MAC table.
- All devices share the same **collision domain**.
- Uses **CSMA/CD** to handle collisions.
- Replaced by switches in modern networks.

- **Half-duplex** operation.

In One Line

A hub broadcasts data to all ports — simple, dumb, and creates collision/broadcast domains.

26. Switch

Definition

A **switch** is a Layer 2 networking device that **forwards frames only to the correct destination port** using a **MAC address table**.

How a Switch Works

1. Frame arrives at port 1 with Src=AA:BB, Dst=CC:DD
2. Switch learns: AA:BB is on port 1 (adds to MAC table)
3. Switch looks up CC:DD in MAC table
4. If found → forwards to that port only (unicast)
5. If not found → floods to all ports except incoming (unknown unicast)

Key Points

- Each port is its own **collision domain** (full-duplex support).
- All ports share one **broadcast domain** (unless VLANs used).
- **MAC address table** (CAM table) stores port-to-MAC mappings.
- More efficient and secure than hubs.

In One Line

A switch intelligently forwards frames to specific ports using a MAC table — eliminates collisions.

27. Router

Definition

A **router** is a Layer 3 networking device that **routes packets between different networks** using **IP addresses** and a **routing table**.

How a Router Works

1. Packet arrives with destination IP
 2. Router checks routing table for best match (longest prefix match)
 3. Forwards packet out the appropriate interface toward destination
 4. Decrements TTL; discards if TTL = 0

Key Points

- Connects different **networks** (LANs, WANs).
- Each interface is a separate **broadcast domain**.
- Uses **routing protocols** (OSPF, BGP, RIP) to build routing table.
- Performs **NAT** to translate private/public IPs.

In One Line

A router routes packets between different networks using IP addresses and a routing table.

28. Hub vs Switch

Feature	Hub	Switch
OSI Layer	Layer 1 (Physical)	Layer 2 (Data Link)
Intelligence	None	MAC address table
Forwarding	Broadcasts to all ports	Forwards to specific port
Collision domain	One for all ports	One per port

Broadcast domain	One	One (unless VLANs)
Speed	Half-duplex	Full-duplex
Efficiency	Low	High
Security	Low	Higher
Used today	Rarely	Yes, universally

In One Line

Hub = broadcast to all; Switch = forward to correct port using MAC table — switches replaced hubs.

29. Switch vs Router

Feature	Switch	Router
OSI Layer	Layer 2	Layer 3
Address used	MAC address	IP address
Connects	Devices in same network	Different networks
Broadcast domain	One (unless VLANs)	Separates broadcast domains
Routing table	No	Yes
NAT	No	Yes
Primary job	Frame switching	Packet routing

In One Line

Switch connects devices within a LAN using MAC; Router connects networks using IP.

30. Broadcast, Unicast & Multicast

Type	Description	Destination
Unicast	One sender → one specific receiver	Specific MAC / IP
Broadcast	One sender → all devices in network	FF:FF:FF:FF:FF:FF (MAC) / 255.255.255.255 (IP)
Multicast	One sender → group of interested receivers	Multicast group address
Anycast	One sender → nearest of a group	Used in IPv6, DNS

In One Line

Unicast=one-to-one; Broadcast=one-to-all; Multicast=one-to-group.

SECTION 4 — IP ADDRESSING & ROUTING

31. IPv4

Definition

IPv4 is the fourth version of the Internet Protocol, using **32-bit addresses** written as **four decimal octets** separated by dots.

Format

192 . 168 . 1 . 1

↑ ↑ ↑ ↑

8 bits 8 bits 8 bits 8 bits = 32 bits total

Range: 0–255 per octet

IPv4 Address Classes

Class	Range	Default Subnet Mask	Use
A	1.0.0.0 – 126.255.255.255	255.0.0.0 (/8)	Large organizations
B	128.0.0.0 – 191.255.255.255	255.255.0.0 (/16)	Medium organizations
C	192.0.0.0 – 223.255.255.255	255.255.255.0 (/24)	Small networks
D	224.0.0.0 – 239.255.255.255	—	Multicast
E	240.0.0.0 – 255.255.255.255	—	Experimental

In One Line

IPv4 is a 32-bit addressing scheme using dotted decimal notation; provides ~4.3 billion addresses.

32. IPv6 Basics

Definition

IPv6 is the sixth version of the Internet Protocol, using **128-bit addresses** to solve IPv4 address exhaustion.

Format

```
2001:0db8:85a3:0000:0000:8a2e:0370:7334
(8 groups of 4 hex digits, separated by colons)
Shortened: 2001:db8:85a3::8a2e:370:7334
```

IPv6 Key Features

- **Huge address space:** $2^{128} \approx 3.4 \times 10^{38}$ addresses.
- **No NAT needed** (enough addresses for all devices).
- **Built-in IPSec** (security).
- **Simplified header** (fixed 40 bytes).
- **No broadcast** → uses multicast and anycast instead.
- **Auto-configuration** (SLAAC — Stateless Address Autoconfiguration).

In One Line

IPv6 is 128-bit addressing solving IPv4 exhaustion; supports auto-configuration and built-in security.

33. Public IP vs Private IP

Feature	Public IP	Private IP
Assigned by	ISP (Internet Service Provider)	Router (DHCP)

Accessible from	Anywhere on Internet	Only within local network
Unique globally	Yes	No (same ranges reused everywhere)
Cost	Paid (limited)	Free (unlimited internally)

Private IP Ranges (RFC 1918)

Class	Private Range
A	10.0.0.0 – 10.255.255.255
B	172.16.0.0 – 172.31.255.255
C	192.168.0.0 – 192.168.255.255

In One Line

Public IPs are globally routable; private IPs are local-only and translated to public via NAT.

34. Static IP vs Dynamic IP

Feature	Static IP	Dynamic IP
Assigned	Manually, fixed permanently	Automatically by DHCP server
Changes	Never	Changes on reconnect / lease expiry

Best for	Servers, printers, DNS	End-user devices (laptops, phones)
Cost	More expensive	Usually included
Management	Manual configuration	Automatic

In One Line

Static IP is manually assigned and fixed; dynamic IP is automatically assigned by DHCP and can change.

35. Loopback Address

Definition

The **loopback address** is a special IP address that **routes traffic back to the same device** without going through any network — used for testing.

Key Points

- **IPv4 loopback:** `127.0.0.1` (entire 127.0.0.0/8 block reserved).
- **IPv6 loopback:** `::1`.
- Hostname: `localhost`.
- Used to test if the network stack on the local machine is working.
- Pinging `127.0.0.1` tests the local TCP/IP stack.

In One Line

`127.0.0.1` (localhost) is the loopback address — traffic sent here stays on the same machine.

36. Subnet Mask

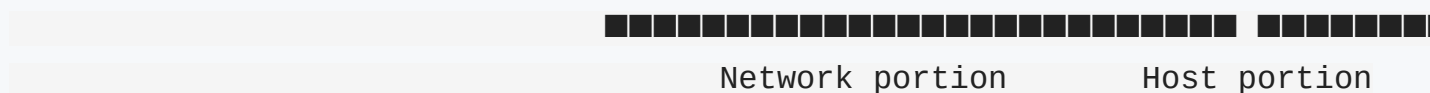
Definition

A **subnet mask** is a 32-bit number that **divides an IP address into network and host portions** by masking bits.

How It Works

IP Address: 192.168.1.100 = 11000000.10101000.00000001.01100100

Subnet Mask: 255.255.255.0 = 11111111.11111111.11111111.00000000



Network: 192.168.1.0

Host: .100

In One Line

Subnet mask identifies which bits of an IP address are the network part and which are the host part.

37. CIDR (Classless Inter-Domain Routing)

Definition

CIDR is a method for **allocating IP addresses** and **routing** that replaces the old class-based system, using a **prefix length** notation.

CIDR Notation

192.168.1.0/24



Common CIDR Values

CIDR	Subnet Mask	Hosts
/8	255.0.0.0	16,777,214

/16	255.255.0.0	65,534
/24	255.255.255.0	254
/25	255.255.255.128	126
/30	255.255.255.252	2

In One Line

CIDR uses /prefix notation to flexibly allocate IP address blocks without class constraints.

38. Subnetting

Definition

Subnetting is the process of **dividing a large network into smaller sub-networks (subnets)** to improve performance and security.

How to Subnet

Example: Divide 192.168.1.0/24 into 4 subnets.

- Need 4 subnets → borrow 2 bits → /26 (256 addresses → $4 \times 64 = 64$ addresses each).

Subnet	Network Address	Broadcast	Usable Range	Hosts
1	192.168.1.0	192.168.1.63	.1 – .62	62
2	192.168.1.64	192.168.1.127	.65 – .126	62
3	192.168.1.128	192.168.1.191	.129 – .190	62

4	192.168.1.192	192.168.1.255	.193 – .254	62
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Formula

Number of subnets = $2^{\text{(borrowed bits)}}$
 Hosts per subnet = $2^{\text{(host bits)}} - 2$ (subtract network + broadcast)
 Block size = 256 - last octet of subnet mask

In One Line

Subnetting divides a network into smaller subnets by borrowing host bits — improves security and reduces broadcast traffic.

39. Network Address & Broadcast Address

Network Address

- **First address** in a subnet.
- Identifies the **subnet itself** — not assignable to any host.
- Example: 192.168.1.0 in a /24 subnet.

Broadcast Address

- **Last address** in a subnet.
- Sends data to **all hosts** in that subnet simultaneously.
- Example: 192.168.1.255 in a /24 subnet.

Usable Hosts = Total addresses – 2 (network + broadcast)

In One Line

Network address = first (identifies subnet); Broadcast = last (sends to all hosts); both are non-assignable.

40. Default Gateway

Definition

A **default gateway** is the **router's IP address** on the local network that devices use to send traffic **destined for outside the local network**.

Key Points

- If the destination IP is **not in the local subnet** → send to default gateway.
- Usually the **first or last usable IP** in a subnet (e.g., 192.168.1.1).
- Without a gateway, a device can only communicate within its own subnet.

In One Line

Default gateway is the local router's IP — all traffic for external networks is sent through it.

41. Routing

Definition

Routing is the process of **selecting the best path** for packets to travel from source to destination across multiple networks.

Routing Process

1. Packet arrives at router with destination IP
2. Router looks up routing table (longest prefix match)
3. Forwards to next-hop router or directly to destination
4. Repeats at each router until destination reached

Key Terms

- **Next hop** → IP address of the next router toward the destination.
- **Metric** → Cost of a route (hop count, bandwidth, delay).
- **TTL (Time to Live)** → Decrement at each hop; packet discarded when TTL=0.

In One Line

Routing selects the best path for packets across networks using a routing table.

42. Routing Table

Definition

A **routing table** is a **database in a router** that stores information about known network paths used to forward packets.

Routing Table Entry

Network	Subnet Mask	Next Hop	Interface	Metric
192.168.1.0	/24	—	eth0	0
10.0.0.0	/8	192.168.1.1	eth1	1
0.0.0.0	/0	203.0.113.1	eth2	10

- **0.0.0.0/0** → Default route (used when no specific match).
- **Longest prefix match** → Most specific route wins.

In One Line

Routing table stores network-to-next-hop mappings; longest prefix match determines the forwarding path.

43. Static vs Dynamic Routing

Feature	Static Routing	Dynamic Routing
Configuration	Manually by admin	Automatically by routing protocols

Updates	Manual	Automatic (protocols exchange info)
Overhead	Low (no protocol traffic)	Higher (routing protocol messages)
Scalability	Poor (hard for large networks)	Good (auto-adapts)
Fault tolerance	None (manual fix required)	Good (reroutes around failures)
Examples	Small networks, specific routes	OSPF, RIP, BGP, EIGRP

In One Line

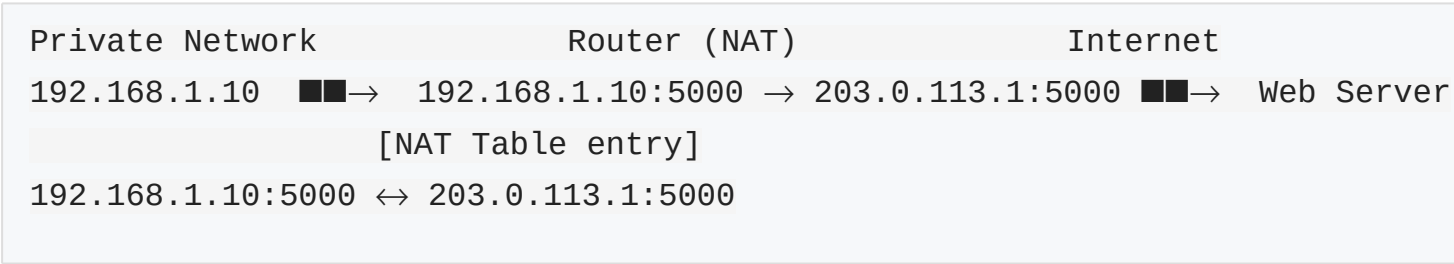
Static routing is manually configured and predictable; dynamic routing auto-updates using routing protocols.

44. NAT (Network Address Translation)

Definition

NAT is a technique where a **router replaces private IP addresses with a public IP** when forwarding packets to the Internet, and reverses the mapping for replies.

How NAT Works



Types of NAT

- **SNAT (Static NAT)** → 1-to-1 mapping (one private to one public).
- **DNAT (Dynamic NAT)** → Many-to-pool mapping.
- **PAT / Masquerading (NAPT)** → Many-to-one; uses port numbers to distinguish connections.

Benefits

- **Conserves IPv4 addresses** → Many devices share one public IP.
- **Security** → Internal IPs hidden from Internet.

In One Line

NAT translates private IPs to a public IP for Internet access, conserving IPv4 addresses.

SECTION 5 — TRANSPORT LAYER

45. TCP (Transmission Control Protocol)

Definition

TCP is a **connection-oriented, reliable transport protocol** that guarantees data delivery in order, without duplicates, and without errors.

Key Features

- **Connection-oriented** → Must establish connection before data transfer (3-way handshake).
- **Reliable** → Acknowledgements, retransmission on loss.
- **Ordered** → Sequence numbers ensure correct order.
- **Error checked** → Checksum on every segment.
- **Flow control** → Sliding window.
- **Congestion control** → Slow start, congestion avoidance.

In One Line

TCP is reliable, ordered, connection-oriented delivery — ensures data arrives correctly, in order.

46. UDP (User Datagram Protocol)

Definition

UDP is a **connectionless, unreliable transport protocol** that sends datagrams with no guarantee of delivery, order, or error recovery.

Key Features

- **Connectionless** → No handshake; send and forget.
- **Unreliable** → No ACKs, no retransmission.
- **No ordering** → Datagrams may arrive out of order.
- **Fast & Low overhead** → Minimal header (8 bytes vs TCP's 20+ bytes).
- **Best for:** Streaming, gaming, DNS, VoIP, DHCP.

In One Line

UDP is fast, connectionless, and unreliable — used when speed matters more than delivery guarantee.

47. TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (ACK + retransmit)	Unreliable (no ACK)

Ordering	Ordered (sequence numbers)	No ordering
Speed	Slower (overhead)	Faster
Header size	20–60 bytes	8 bytes
Flow control	Yes (sliding window)	No
Congestion control	Yes	No
Use cases	HTTP, FTP, SMTP, SSH	DNS, DHCP, VoIP, Video, Gaming
Error recovery	Yes	No (app handles if needed)

In One Line

TCP = reliable but slow; UDP = fast but unreliable — choose based on application need.

48. Port Numbers

Definition

Port numbers identify specific **applications or services** on a host. Combined with an IP address, they form a socket.

Key Well-Known Ports

Port	Protocol	Service
20, 21	TCP	FTP (data/control)

22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	TCP/UDP	DNS
67/68	UDP	DHCP
80	TCP	HTTP
110	TCP	POP3
143	TCP	IMAP
443	TCP	HTTPS
3306	TCP	MySQL

In One Line

Port numbers identify services on a host; well-known ports (0–1023) are reserved for standard services.

49. Connection-Oriented vs Connectionless

Feature	Connection-Oriented (TCP)	Connectionless (UDP)
Setup	Handshake required	No setup

State	Maintains connection state	No state
Reliability	Guaranteed delivery	Best-effort
Overhead	High	Low
Examples	TCP, ATM	UDP, IP

In One Line

Connection-oriented (TCP) establishes a dedicated path before transfer; connectionless (UDP) just sends.

50. Flow Control & Congestion Control

Flow Control

- **Purpose:** Prevents the **sender from overwhelming the receiver's buffer**.
- **Mechanism:** Receiver advertises a **receive window (rwnd)** — max data it can accept.
- Sender limits unacknowledged data to $\leq$ rwnd.

Congestion Control

- **Purpose:** Prevents the **sender from overwhelming the network** (routers/links).
- **Mechanism:** TCP maintains a **congestion window (cwnd)**.
- **Slow Start** → cwnd starts small; doubles every RTT.
- **Congestion Avoidance** → After threshold: cwnd increases linearly.
- **Fast Retransmit** → 3 duplicate ACKs → retransmit without waiting for timeout.

In One Line

Flow control protects the receiver's buffer; congestion control protects the network — both use window sizes.

51. Sliding Window

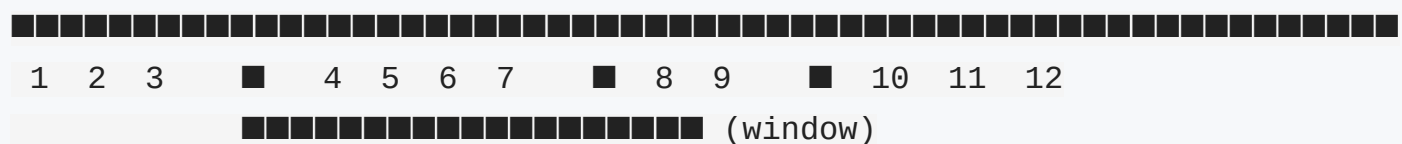
Definition

Sliding window is a **flow control mechanism** that allows the sender to have **multiple unacknowledged packets** in transit simultaneously, improving throughput.

How It Works

Sender window size = 4 (can send 4 packets before waiting for ACK)

Sent & ACKed | Sent, not ACKed | Can send | Cannot send yet



As ACK arrives, window "slides" forward

Key Points

- **Window size** determines throughput.
- **Go-Back-N** → On error, retransmit from error onwards.
- **Selective Repeat** → Retransmit only the lost packet.

In One Line

Sliding window allows multiple packets in flight; window slides forward as ACKs arrive — maximizes throughput.

...

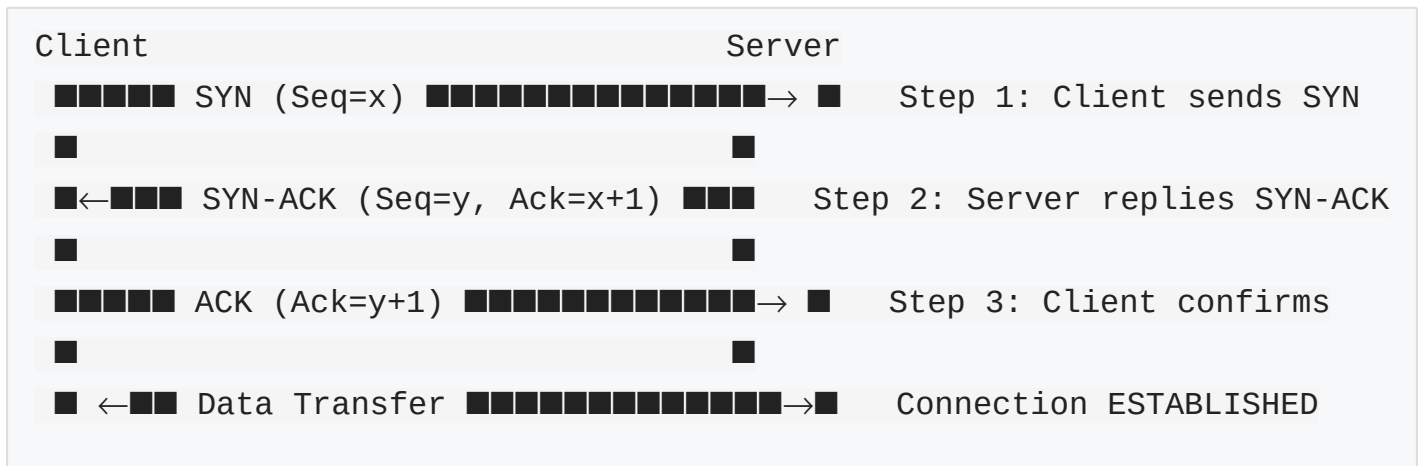
SECTION 6 — TCP CONNECTION & RELIABILITY

52. TCP 3-Way Handshake

Definition

The **TCP 3-Way Handshake** is the process TCP uses to **establish a connection** between client and server before data transfer begins.

Steps



Purpose of Each Step

- **SYN** → Client requests connection; sends initial sequence number.
- **SYN-ACK** → Server acknowledges; sends its own sequence number.
- **ACK** → Client acknowledges server's sequence number.

In One Line

TCP 3-Way Handshake (SYN → SYN-ACK → ACK) establishes a reliable connection before data transfer.

53. SYN, SYN-ACK, ACK

Flag	Meaning	Direction	Purpose
SYN	Synchronize	Client → Server	Initiate connection; share initial seq number

TCP closes with 4 steps (FIN → ACK → FIN → ACK) since each side terminates independently.

55. Sequence Number & Acknowledgement Number

Sequence Number

- A **32-bit number** in the TCP header that **identifies the byte position** of the first byte in the segment.
- Ensures data is reassembled in the correct order.

Acknowledgement Number

- Tells the sender the **next byte expected** from it.
- ACK number = last received byte + 1 = "send me from this byte next."

Example

```
Sender sends Seq=100, 10 bytes of data (bytes 100-109)
Receiver replies: ACK=110 (next expected byte)
Sender now knows bytes 100-109 were received correctly
```

In One Line

Sequence number labels data bytes; ACK number tells sender the next expected byte — together they ensure ordered, reliable delivery.

56. Retransmission & Timeout

Retransmission

- If an ACK is **not received** within a timeout period, TCP **retransmits** the unacknowledged segment.
- Also triggered by **3 duplicate ACKs** (fast retransmit).

Timeout (RTO — Retransmission Timeout)

- Calculated dynamically based on **RTT (Round Trip Time)**.
- $RT0 = EstimatedRTT + 4 \times DevRTT$ (Jacobson/Karels algorithm).
- **Too short** → Unnecessary retransmissions.
- **Too long** → Slow recovery from loss.

In One Line

Retransmission recovers lost segments — triggered by timeout or 3 duplicate ACKs.

57. TCP Connection States

State	Description
CLOSED	No connection; initial state
LISTEN	Server waiting for incoming connection requests
SYN_SENT	Client sent SYN; waiting for SYN-ACK
SYN_RECEIVED	Server received SYN; sent SYN-ACK
ESTABLISHED	Connection open; data transfer in progress
FIN_WAIT_1	Client sent FIN; waiting for ACK
FIN_WAIT_2	Client received ACK; waiting for server's FIN
CLOSE_WAIT	Server received FIN; waiting for app to close

TIME_WAIT	Client sent final ACK; waiting 2×MSL before closing
LAST_ACK	Server waiting for final ACK from client

In One Line

TCP goes through multiple states: LISTEN → ESTABLISHED (data) → FIN_WAIT → TIME_WAIT → CLOSED.

58. TIME_WAIT & CLOSE_WAIT

TIME_WAIT

- State entered by the **active closer** (usually client) after sending final ACK.
- Lasts **2 × MSL (Maximum Segment Lifetime)** ≈ 60–120 seconds.
- **Why:** Ensures the final ACK reached server; allows old duplicate packets to expire.

CLOSE_WAIT

- State entered by **passive closer** (usually server) after receiving FIN.
- Waits for the **application to call close()**.
- If many sockets stuck in CLOSE_WAIT → application bug (not calling close).

In One Line

TIME_WAIT is the client's safety wait after closing; CLOSE_WAIT is the server waiting for the app to close its socket.

59. TCP Flow Control

Definition

TCP Flow Control ensures the sender doesn't **overwhelm the receiver's buffer** by limiting how much data can be in transit.

Mechanism: Receive Window (rwnd)

- Receiver advertises **rwnd** (free space in its buffer) in every ACK.
- Sender limits unacknowledged data to $\leq$ rwnd.
- If rwnd = 0 $\rightarrow$ sender stops (sends probes to check when receiver is ready).

Sender ■■ $\rightarrow$ up to rwnd bytes unacknowledged ■■ $\rightarrow$ Receiver
 $\leftarrow$ ■■ ACK with new rwnd value ■■■■■■

In One Line

TCP flow control uses the receiver's advertised window (rwnd) to prevent buffer overflow at the receiver.

60. TCP Congestion Control

Definition

TCP Congestion Control prevents the sender from **overloading the network** by dynamically adjusting the sending rate based on network conditions.

Phases

1. Slow Start:
 - cwnd = 1 MSS
 - Double cwnd each RTT (exponential growth)
 - Until cwnd reaches ssthresh
2. Congestion Avoidance:
 - Increase cwnd by 1 MSS per RTT (linear)
 - Until packet loss detected
3. On Packet Loss:
 - (a) Timeout $\rightarrow$ ssthresh = cwnd/2; cwnd = 1 (restart slow start)
 - (b) 3 dup ACKs (Fast Recovery) $\rightarrow$ ssthresh = cwnd/2; cwnd = ssthresh

In One Line

TCP congestion control uses slow start (exponential) then congestion avoidance (linear) to find the network's capacity without causing collapse.

SECTION 7 — APPLICATION-LEVEL NETWORKING

61. DNS (Domain Name System)

Definition

DNS is a **hierarchical distributed naming system** that translates **human-readable domain names** (e.g., google.com) to **IP addresses** (e.g., 142.250.182.46).

How DNS Resolution Works

```
Browser: "What is the IP of www.google.com?"  
  ↓  
1. Check local DNS cache  
2. Ask Recursive Resolver (ISP's DNS)  
3. Resolver asks Root DNS Server (knows TLD servers)  
4. Root replies: "Ask .com TLD server"  
5. Resolver asks .com TLD server  
6. TLD replies: "Ask google.com's authoritative NS"  
7. Resolver asks google.com's authoritative server  
8. Gets IP → returns to browser → browser connects
```

Key DNS Record Types

Record	Purpose
A	Domain → IPv4 address

AAAA	Domain → IPv6 address
CNAME	Alias → canonical domain name
MX	Mail exchange server for domain
NS	Authoritative name servers
PTR	Reverse DNS (IP → domain)
TXT	Text records (SPF, DKIM, verification)

In One Line

DNS translates domain names to IPs; resolution goes from local cache → recursive resolver → root → TLD → authoritative DNS.

62. DHCP (Dynamic Host Configuration Protocol)

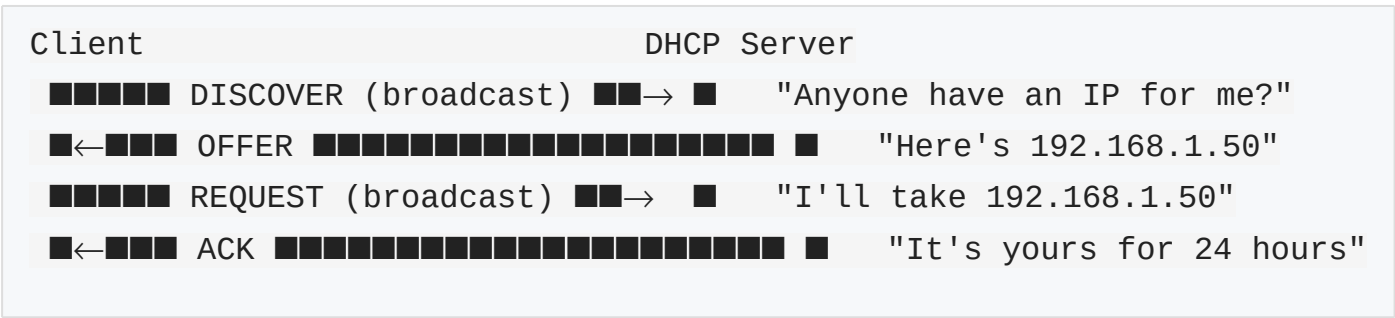
Definition

DHCP is a network protocol that **automatically assigns IP addresses** and other network configuration to devices when they join a network.

What DHCP Assigns

- IP address, Subnet mask, Default gateway, DNS server addresses, Lease time.

DHCP Process (DORA)



- Discover → Offer → Request → Acknowledge

In One Line

DHCP automatically assigns IP configuration via DORA — Discover, Offer, Request, Acknowledge.

63. HTTP (HyperText Transfer Protocol)

Definition

HTTP is an **application-layer protocol** used for **transferring web pages** between a client (browser) and a web server.

Key Points

- **Stateless** → Each request is independent; server remembers nothing.
- **Request-Response** model.
- Default port: **80**.
- Uses **TCP** as transport.
- Text-based protocol (human-readable headers).

HTTP Request Structure

```
GET /index.html HTTP/1.1
Host: www.example.com
Accept: text/html
User-Agent: Mozilla/5.0
```

HTTP Response Structure

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 1024
```

```
<html>...</html>
```

In One Line

HTTP is the stateless request-response protocol for web communication on port 80.

64. HTTPS

Definition

HTTPS (HTTP Secure) is HTTP with **SSL/TLS encryption** — ensuring data confidentiality, integrity, and server authentication.

How HTTPS Works

1. Client connects to server on port 443
2. TLS Handshake: negotiate cipher suite; server sends certificate
3. Client verifies certificate with CA
4. Session key established
5. All HTTP data encrypted with session key

HTTP vs HTTPS

Feature	HTTP	HTTPS
Port	80	443
Encryption	No	Yes (TLS)
Security	None	Encrypted + Authenticated
Certificate	Not required	Required (from CA)

URL prefix	http://	https://
------------	---------	----------

In One Line

HTTPS = HTTP + TLS encryption; provides confidentiality, integrity, and authentication on port 443.

65. HTTP Methods

Method	Purpose	Has Body?	Idempotent?	Safe?
GET	Retrieve a resource	No	Yes	Yes
POST	Create a new resource	Yes	No	No
PUT	Replace an entire resource	Yes	Yes	No
PATCH	Partially update a resource	Yes	No	No
DELETE	Delete a resource	Optional	Yes	No
HEAD	Like GET but no body in response	No	Yes	Yes

OPTIONS	List supported methods	No	Yes	Yes
----------------	------------------------	----	-----	-----

- **Idempotent** → Calling multiple times gives same result.
- **Safe** → Doesn't modify server state.

In One Line

GET=read, POST=create, PUT=replace, PATCH=update, DELETE=remove — the standard HTTP methods.

66. HTTP Status Codes

Range	Category	Common Codes
1xx	Informational	100 Continue
2xx	Success	200 OK, 201 Created, 204 No Content
3xx	Redirection	301 Moved Permanently, 302 Found, 304 Not Modified
4xx	Client Error	400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 429 Too Many Requests
5xx	Server Error	500 Internal Server Error, 502 Bad Gateway, 503 Service Unavailable

In One Line

2xx=success, 3xx=redirect, 4xx=client error, 5xx=server error — five classes of HTTP status codes.

67. HTTP Headers

Definition

HTTP headers are **key-value pairs** sent in requests and responses, providing metadata about the message.

Common Request Headers

Header	Purpose
Host	Target server domain
Authorization	Credentials (e.g., Bearer token)
Content-Type	Media type of request body
Accept	Media types client accepts
Cookie	Session cookies sent to server
User-Agent	Client browser/OS info

Common Response Headers

Header	Purpose
Content-Type	Media type of response body

Set-Cookie	Server sets a cookie on client
Cache-Control	Caching instructions
Location	Redirect target URL (3xx)
WWW-Authenticate	Auth challenge (401)

In One Line

HTTP headers carry metadata about requests and responses — authentication, content type, caching, cookies.

68. Cookies & Sessions

Cookies

- **Small pieces of data** stored by the **browser**, sent with every request to the same origin.
- Set by server via Set-Cookie response header.
- Used for: session management, personalization, tracking.
- **Persistent cookies** → Have an expiry date.
- **Session cookies** → Deleted when browser closes.

Sessions

- **Server-side storage** of user state.
- Browser stores only a **Session ID** (in a cookie).
- Server uses Session ID to look up full session data.

Cookie vs Session

Feature	Cookie	Session
---------	--------	---------

Storage	Client (browser)	Server
Security	Less secure (client-side)	More secure (server-side)
Size limit	~4KB	Larger (server storage)
Expiry	Can be long-lived	Usually expires on close

In One Line

Cookies store data on the browser; sessions store data on the server with a cookie holding only the session ID.

69. HTTP/1.1 vs HTTP/2 vs HTTP/3

Feature	HTTP/1.1	HTTP/2	HTTP/3
Year	1997	2015	2022
Transport	TCP	TCP	QUIC (UDP-based)
Multiplexing	No (one request/connection)	Yes (multiple streams)	Yes
Header compression	No	HPACK compression	QPACK
Server push	No	Yes	Yes
HOL blocking	Yes (TCP)	Yes (TCP, not HTTP)	No (QUIC)

TLS	Optional	Required (de facto)	Built-in (required)
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Key Improvements

- **HTTP/2** → Multiplexing eliminates HTTP-level head-of-line blocking.
- **HTTP/3** → Uses QUIC (UDP) to eliminate TCP head-of-line blocking; faster connection setup.

In One Line

HTTP/1.1=one-at-a-time; HTTP/2=multiplexed over TCP; HTTP/3=multiplexed over QUIC(UDP) with built-in TLS.

SECTION 8 — NETWORK SECURITY BASICS

70. HTTP vs HTTPS

(See Topic 64 for full comparison.)

Feature	HTTP	HTTPS
Encryption	No	Yes (TLS)
Port	80	443
Data visible to ISP	Yes	No
MITM attack risk	High	Low
Certificate	Not required	Required

In One Line

HTTP is plaintext; HTTPS is encrypted HTTP using TLS — always use HTTPS for sensitive data.

71. SSL/TLS Basics

Definition

- **SSL (Secure Sockets Layer)** → Original protocol; now deprecated (SSL 3.0 vulnerable).
- **TLS (Transport Layer Security)** → Modern, secure replacement for SSL.
- Provides: **Encryption** (confidentiality), **Integrity** (tamper detection), **Authentication** (verify server identity).

TLS Handshake (Simplified)

1. Client Hello → TLS version, supported cipher suites, random number
2. Server Hello → chosen cipher, server's SSL certificate, random number
3. Client verifies certificate with CA
4. Key Exchange → agree on session key (using asymmetric crypto)
5. Session established → all data encrypted with symmetric session key

TLS Versions

Version	Status
SSL 3.0	Deprecated (insecure)
TLS 1.0, 1.1	Deprecated
TLS 1.2	Widely used
TLS 1.3	Current standard; faster, more secure

In One Line

TLS provides encrypted, authenticated communication — used by HTTPS; consists of handshake + encrypted data transfer.

72. Symmetric Encryption

Definition

Symmetric encryption uses the **same key for both encryption and decryption**.

Key Points

- **Fast** → Efficient for bulk data encryption.
- **Problem:** Key distribution — how to securely share the key?
- Used for: encrypting actual data after TLS handshake.
- Examples: **AES (Advanced Encryption Standard)**, DES, 3DES, ChaCha20.

Plaintext ■■[Key]■■→ Ciphertext ■■[Key]■■→ Plaintext
Encrypt Decrypt
(same key both ways)

In One Line

Symmetric encryption uses one key for encrypt + decrypt — fast but requires secure key sharing.

73. Asymmetric Encryption

Definition

Asymmetric encryption uses a **key pair** — **public key (encrypt)** and **private key (decrypt)** — mathematically linked but computationally infeasible to derive one from the other.

Key Points

- **Public key** → Can be shared openly; used to encrypt.
- **Private key** → Kept secret by owner; used to decrypt.
- **Slow** → Not used for bulk data; used for key exchange and signatures.
- Examples: **RSA**, ECC, Diffie-Hellman.

Sender: Plaintext [Public Key] → Ciphertext → Receiver
Receiver: Ciphertext [Private Key] → Plaintext

Use Cases

- **TLS key exchange** → Agree on symmetric key securely.
- **Digital signatures** → Sign with private key; verify with public key.
- **SSH** → Authenticate without passwords.

In One Line

Asymmetric encryption uses public key to encrypt, private key to decrypt — secure key exchange without sharing a secret.

74. Public Key & Private Key

Feature	Public Key	Private Key
Sharing	Share with anyone	Never share; keep secret
Used for	Encryption, verifying signatures	Decryption, creating signatures
Generated	Together as a pair	Together as a pair
If compromised	Can be revoked; get new cert	All security is lost — regenerate

Definition

A **Certificate Authority (CA)** is a **trusted third-party organization** that **issues, validates, and revokes** digital certificates.

How It Works

1. Website owner creates key pair; sends Certificate Signing Request (CSR)
2. CA verifies domain ownership
3. CA signs certificate with its private key
4. Browser trusts CAs in its "root store" (pre-installed)
5. Browser verifies certificate signature using CA's public key

CA Hierarchy

```
Root CA (most trusted, offline)
  ↓
Intermediate CA (signs end-entity certs)
  ↓
End-Entity Certificate (your website cert)
```

Well-Known CAs

DigiCert, Let's Encrypt (free), GlobalSign, Comodo, Sectigo.

In One Line

CA is the trusted authority that signs digital certificates — browsers trust CAs to verify website identity.

77. Firewall

Definition

A **firewall** is a **network security device (hardware or software)** that **monitors and controls incoming and outgoing network traffic** based on predefined security rules.

Types of Firewalls

Type	Description
Packet Filter	Inspects IP/port headers; stateless; fast but limited
Stateful Firewall	Tracks connection state; smarter filtering
Application Firewall (WAF)	Inspects application-layer data (HTTP content)
Next-Gen Firewall (NGFW)	DPI + IDS/IPS + app awareness + user identity

Firewall Rules

```
Allow TCP from 192.168.1.0/24 to any port 80
Allow TCP from 192.168.1.0/24 to any port 443
DENY all other inbound traffic
```

In One Line

A firewall filters traffic by rules — allowing legitimate traffic and blocking threats.

78. VPN (Virtual Private Network)

Definition

A **VPN** creates an **encrypted tunnel** over a public network (Internet), allowing remote users or offices to communicate as if on a **private network**.

How VPN Works

```
User Device [Encrypted Tunnel] → VPN Server → Internet
(public Internet used, but traffic is encrypted)
```

Types

Type	Use Case
Remote Access VPN	Individual user connects to corporate network from home
Site-to-Site VPN	Connects two office networks permanently
SSL/TLS VPN	Browser-based; no special client needed

VPN Protocols

- **OpenVPN** → Open-source; very secure.
- **WireGuard** → Modern, fast, minimal.
- **IPSec** → Standard; used in site-to-site.
- **L2TP/IPSec** → L2TP tunneling + IPSec encryption.

Benefits

- **Privacy** → ISP and others can't see traffic contents.
- **Security** → Encrypted, even on public Wi-Fi.
- **Access** → Bypass geo-restrictions; access corporate resources.

In One Line

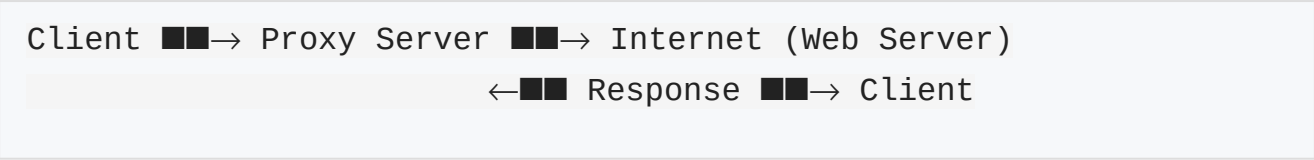
VPN creates an encrypted tunnel over the Internet — provides privacy, security, and remote access to private networks.

79. Proxy

Definition

A **proxy server** is an **intermediary server** that sits between clients and the Internet, forwarding requests on behalf of clients.

How It Works



Types of Proxies

Type	Description
Forward Proxy	Client-side; hides client from Internet; caching, filtering
Reverse Proxy	Server-side; hides server from Internet; load balancing, SSL termination
Transparent Proxy	Client doesn't know it's proxied; used by ISPs for caching
SOCKS Proxy	Works at socket level; handles any protocol

Proxy vs VPN

Feature	Proxy	VPN
Encryption	Usually no	Yes
OS-level	No (app-specific)	Yes (all traffic)
Speed	Faster	Slightly slower

Privacy	Partial	Full
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In One Line

A proxy forwards requests between client and server — forward proxy hides clients; reverse proxy protects servers.

QUICK REVISION TABLE — ALL TOPICS

#	Topic	Key Point
1	Computer Network	Devices connected to share data/resources
2	Internet	Global network of networks using TCP/IP
3	Client and Server	Client requests; server responds
4	Network Protocol	Rules for communication format and timing
5	Packet	Data broken into header + payload chunks
6	Port	Logical endpoint identifying a service (0–65535)
7	Socket	IP + Port + Protocol = communication endpoint

8	IP Address	Unique numerical ID for network device
9	MAC Address	48-bit hardware NIC address for LAN delivery
10	Bandwidth	Max data capacity; throughput is actual rate
11	Latency	Propagation + Transmission + Processing + Queuing delay
12	OSI Model	7 layers: Physical→DataLink→Network→Transp
13	Physical Layer	Layer 1; transmits raw bits; hubs, cables
14	Data Link Layer	Layer 2; frames, MAC, CRC, switches
15	Network Layer	Layer 3; IP, routing, packets, routers
16	Transport Layer	Layer 4; TCP/UDP, ports, end-to-end delivery
17	Application Layer	Layer 7; HTTP, DNS, FTP, SMTP
18	TCP/IP Model	4 layers: Network Access→Internet→Transport→Applica

19	OSI vs TCP/IP	OSI=7 layer reference; TCP/IP=4 layer practical
20	Encapsulation	Add headers at each layer going down
21	MAC Address	First 3 bytes=manufacturer; last 3=device; ARP maps IP→MAC
22	Ethernet	Wired LAN standard; frames; CSMA/CD; IEEE 802.3
23	Frame	Layer 2 PDU; src/dst MAC + payload + FCS
24	ARP	Resolves IP→MAC via broadcast request + unicast reply
25	Hub	Layer 1; broadcasts to all; one collision domain
26	Switch	Layer 2; MAC table; forwards to specific port; one per collision domain
27	Router	Layer 3; routes between networks using IP and routing table
28	Hub vs Switch	Hub=broadcast,dumb; Switch=forward,smart

29	Switch vs Router	Switch=MAC/LAN; Router=IP/between networks
30	Broadcast/Unicast/Multicast	One-to-all / One-to-one / One-to-group
31	IPv4	32-bit, dotted decimal, 4.3B addresses, 5 classes
32	IPv6	128-bit, hex, 3.4×10^3 addresses, no NAT needed
33	Public vs Private IP	Public=globally routable; private=local, RFC 1918 ranges
34	Static vs Dynamic IP	Static=manual, fixed; Dynamic=DHCP, changes
35	Loopback	127.0.0.1 (localhost); tests local TCP/IP stack
36	Subnet Mask	Separates network and host bits of an IP
37	CIDR	/prefix notation; replaces classful addressing
38	Subnetting	Borrow host bits; 2^n subnets; $2^h - 2$ hosts

39	Network/Broadcast Address	First=network; Last=broadcast; both non-assignable
40	Default Gateway	Router IP; used for traffic outside local subnet
41	Routing	Select best path for packets across networks
42	Routing Table	Network→next-hop mappings; longest prefix match
43	Static vs Dynamic Routing	Static=manual; Dynamic=auto via OSPF/BGP/RIP
44	NAT	Translates private→public IP; PAT uses ports for many-to-one
45	TCP	Reliable, ordered, connection-oriented; ACK + retransmit
46	UDP	Fast, connectionless, unreliable; 8-byte header
47	TCP vs UDP	TCP=reliable,slow; UDP=fast,unreliable
48	Port Numbers	0–1023=well-known; 1024–49151=registered; 49152+=ephemeral

49	Connection vs Connectionless	TCP=handshake,state; UDP=no setup,stateless
50	Flow/Congestion Control	Flow=protect receiver buffer; Congestion=protect network
51	Sliding Window	Multiple packets in flight; window slides on ACK
52	3-Way Handshake	SYN→SYN-ACK→ACK establishes TCP connection
53	SYN/SYN-ACK/ACK	Start/Accept/Confirm TCP connection flags
54	4-Way Termination	FIN→ACK→FIN→ACK closes TCP connection
55	Seq/ACK Number	Seq=byte position; ACK=next expected byte
56	Retransmission	Triggered by timeout or 3 dup ACKs
57	TCP States	LISTEN→ESTABLISHED→FIN_WAIT
58	TIME_WAIT/CLOSE_WAIT	TIME_WAIT=client waits 2×MSL; CLOSE_WAIT=server waits for app

59	TCP Flow Control	rwnd limits sender to receiver's buffer capacity
60	TCP Congestion Control	Slow start (exponential) → congestion avoidance (linear)
61	DNS	Translates domain names to IPs; DORA: Root→TLD→Authoritative
62	DHCP	Auto IP assignment via DORA (Discover→Offer→Request→ACK)
63	HTTP	Stateless request-response web protocol; port 80
64	HTTPS	HTTP + TLS encryption; port 443; certificate required
65	HTTP Methods	GET=read, POST=create, PUT=replace, PATCH=update, DELETE=remove
66	HTTP Status Codes	2xx=OK, 3xx=redirect, 4xx=client error, 5xx=server error

67	HTTP Headers	Metadata in requests/responses; Content-Type, Authorization, Cookie
68	Cookies & Sessions	Cookie=browser storage; Session=server storage with cookie ID
69	HTTP/1.1 vs /2 vs /3	1.1=serial; 2=multiplexed/TCP; 3=multiplexed/QUIC(UDP)
70	HTTP vs HTTPS	HTTP=plaintext port 80; HTTPS=encrypted port 443
71	SSL/TLS	TLS secures connections: handshake→session key→encrypted data
72	Symmetric Encryption	One key for both encrypt+decrypt; AES; fast
73	Asymmetric Encryption	Public=encrypt; Private=decrypt; RSA; slow; key exchange
74	Public/Private Key	Public=shared freely; Private=never share; used for crypto+signing

75	Digital Certificate	Binds public key to identity; signed by CA (X.509)
76	Certificate Authority	Trusted org that signs/issues/revokes certificates
77	Firewall	Filters traffic by rules; packet/stateful/application/NGFW
78	VPN	Encrypted tunnel over Internet; remote access + privacy
79	Proxy	Intermediary; forward=hide client; reverse=protect server

Notes prepared for: College Exams | Placement Preparation | Technical Interviews | Quick Revision